LAB 01

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

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Github Link: https://github.com/Harshita2011/Advanced_Predictive_Analytics_LAB

Contents

1. Executive Summary

This report compares seven regression models for house price prediction on three public datasets that differ considerably in size and difficulty: **Ames Housing** (1,460 individual home sales in Ames, Iowa; target `SalePrice` in USD), **California Housing** (20,640 census block groups from the 1990 U.S. Census; target median district value in USD 100,000s), and the **UCI Real Estate Valuation** dataset (414 transactions in New Taipei City, Taiwan; target price per unit area in 10,000 NTD per ping). The models — Linear Regression, Ridge, Lasso, Elastic Net, Decision Tree, Random Forest, and Gradient Boosting — all ran through the same leakage-safe scikit-learn pipeline. Selection was based on 5-fold cross-validation on the training split only, and the held-out 20% test split was touched once, at the end.

Ensembles generalized best on every dataset, though the margin varied. On Ames, the selected **Gradient Boosting** model reached a test MAE of **$16,556**, RMSE of **$26,504**, and RZ of **0.908** — 69.8% below the RMSE of a mean-prediction baseline. On California, **Random Forest** reached MAE 0.327, RMSE 0.505 (USD 100k), and RZ 0.805 (−55.9%); on UCI, it reached MAE 3.913, RMSE 5.669, and RZ 0.808 (−56.2%). The variables carrying most of the signal were living area, overall quality, and neighborhood on Ames; median income and location on California; and MRT-station distance and convenience-store count on UCI. The linear baseline tells its own story: on Ames it stayed close to the ensembles (test RZ 0.887), but on California it stalled at RZ 0.576 against 0.805, which is a fair measure of how much nonlinear structure a straight-line model leaves on the table there.

The models are usable as decision-support estimates within the markets and time periods they were trained on. They should **not** be used for mortgage underwriting, taxation, or other high-stakes decisions without current local data, uncertainty estimates, segment-level validation, and human review.

2. Business Problem

The practical problem is to estimate the market value of a residential property from things that can be measured about it: size, age, quality, location, and nearby amenities. Framed as supervised regression, the goal is to learn a mapping $\hat{y} = f(x)$ from property attributes to a continuous price target, and to judge that mapping only on observations the model never saw during fitting.

2.1 Stakeholders and error costs

- **Buyers and sellers** use the estimates for price discovery and negotiation, where a large error can distort an affordability decision.
- **Banks and mortgage firms** run valuation checks against collateral risk. Systematic over-valuation increases credit exposure, which is why RMSE — it penalizes large errors more heavily — is treated as the primary metric alongside MAE.
- **Developers and investors** screen sites and portfolios with these estimates; location-dependent bias can misallocate capital.
- **Public agencies** need transparent, consistent mass appraisal, which is one reason to keep an interpretable baseline in the comparison.

2.2 Prediction units and success criteria

The three targets are different quantities, and the report never compares them directly. Ames predicts an **individual sale price in USD**. California predicts a **median block-group value** in USD 100,000s — not the price of an individual house. UCI predicts **price per unit area** in 10,000 NTD per ping, not the total transaction price. A model counts as successful here if it (i) clearly beats a mean-prediction baseline on cross-validation and test RMSE/MAE, (ii) shows a small gap between cross-validation and test performance, (iii) leaves no unexplained severe residual pattern and no sign of leakage, and (iv) reports errors in currency units that mean something to a practitioner.

3. Dataset Description

Table 1 summarizes the three datasets. They were chosen to escalate in difficulty: UCI is small and clean, California is large and numeric, and Ames — the main subject of the extended study — is a realistic high-dimensional mix of numerical and categorical variables.

Table 1. Dataset summary

Dataset	Rows	Predictors	Types	Target and unit	Source / coverage
Ames Housing	1,460	80 (37 num, 43 cat, excl. Id)	Mixed	SalePrice, USD	Ames, Iowa residential sales, 2006–2010 (De Cock, 2011)[1]
California Housing	20,640	8 numeric	Numeric	MedHouseVal, USD 100k (block-group median)	1990 U.S. Census, via scikit-learn[2][3]
UCI Real Estate	414	6 numeric	Numeric	Price per unit area, 10k NTD/ping	New Taipei City, Taiwan, 2012–2013 (UCI ID 477)[4]

The data-quality audit turned up a few things worth recording. Ames has no duplicate rows, but 19 of its 81 columns contain missing values, and most of the heavily missing ones are absence codes rather than unknowns: `PoolQC` is missing for 99.5% of rows because those houses have no pool, and `Alley` (93.8%) means no alley access. These were kept and encoded as an explicit category instead of being imputed away. The Ames target is strongly right-skewed (skewness 1.88; mean \$180,921 against a median of \$163,000; range \$34,900–\$755,000). California has no missing values at all, but two caveats apply: the target is right-censored (median values capped at \$500,001), and each row is a block group rather than a house. UCI is the simplest of the three — 414 rows, six numeric predictors, nothing missing.

4. Methodology

4.1 Experimental protocol

The same protocol was applied to all three datasets, so differences in results reflect the data and the models rather than the setup:

- **Holdout design.** An 80/20 random train–test split (`random_state=42`) was made before any learned preprocessing, giving 1,168/292 observations (Ames), 16,512/4,128 (California), and 331/83 (UCI).
- **Leakage-safe preprocessing.** Numeric features went through median imputation and standard scaling; categorical features through most-frequent imputation and one-hot encoding (`handle_unknown='ignore'`). Everything sits in a `ColumnTransformer` inside a scikit-learn `Pipeline` , and every transformer is fit on training folds only.
- **Model selection.** Candidates were compared by 5-fold cross-validation (shuffled, seed 42) on the training split. The test set was used once, for the final evaluation.
- **Hyperparameter tuning.** Ridge α was tuned with `GridSearchCV` over 13 log-spaced values in [10ˉ3, 103], scored by negative cross-validation RMSE.
- **Reproducibility.** Seed 42 is used for every stochastic step. Package versions are listed in Appendix A, final pipelines are serialized with `joblib` , and the raw data files were never edited.

4.2 Models and theory anchor

Ordinary Least Squares (OLS) estimates coefficients by minimizing the residual sum of squares:

$$\hat{\beta} = arg\ min_{\beta}\ \Sigma_i\ (y_i - \beta_0 - \beta^T x_i)^2 = (X^T X)^{-1} X^T y \tag{1}$$

Multiple Linear Regression is the baseline everything else is measured against, and it is the model whose coefficients can actually be read. Ridge adds an L2 penalty λΣβ_jZ to the OLS objective, which keeps correlated coefficients from blowing up; Lasso uses an L1 penalty λΣ|β_j| and can push coefficients to exactly zero; Elastic Net sits between the two. The tree models work differently. A single Decision Tree can follow nonlinear patterns and interactions but has high variance; Random Forest averages many trees to bring that variance down; Gradient Boosting builds trees one at a time, each correcting the errors of the last. Accuracy is reported as MAE and RMSE in the original target units, plus the coefficient of determination:

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2 \tag{2}$$

4.3 Naive baselines

Before fitting anything, a mean-prediction baseline sets the bar: if a model cannot beat the training mean, nothing has been learned. The test-set baseline RMSE is $87,619 for Ames, 1.145 (USD 100k) for California, and 12.95 (10k NTD/ping) for UCI — the last one derived exactly from the reported Linear Regression RZ. On Ames and California every fitted model beats its baseline by a wide margin. On UCI the same is true for all models except the Decision Tree, with reductions of roughly 37–56%.

5. Exploratory Data Analysis and Feature Engineering

All exploratory analysis was done on the training data; the test set played no part in any feature-engineering decision. Each dataset had something different to say, and those findings fed directly into the modeling choices below.

5.1 Ames Housing

- **Right-skewed target.** Figure 1 shows the long right tail of sale prices. This is why RMSE matters here — a few expensive mistakes hurt — and it is the reason a log-target variant is considered in Section 10.
- **Informative missingness.** Figure 2 shows that the most-missing features are absence codes (no pool, no alley, no fence, no fireplace). Dropping the affected rows would throw away roughly 90% of the data, so these features were instead given an explicit "None" category.
- **Correlated size and quality signals.** Figure 3 ranks the strongest numerical correlates of price. OverallQual (r = 0.79) and GrLivArea (r = 0.71) lead by a clear margin, and several others — garage size and count, basement and first-floor area — largely duplicate each other. Redundancy of this kind is exactly what regularization is for.

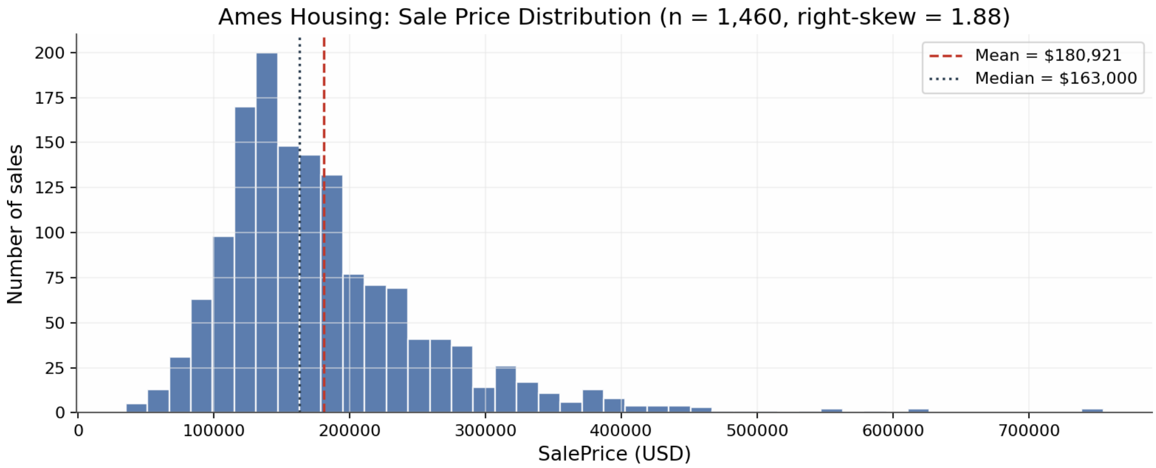

Figure 1. Ames Housing: sale-price distribution (n = 1,460). Right skew 1.88; mean exceeds median by ~$18k.

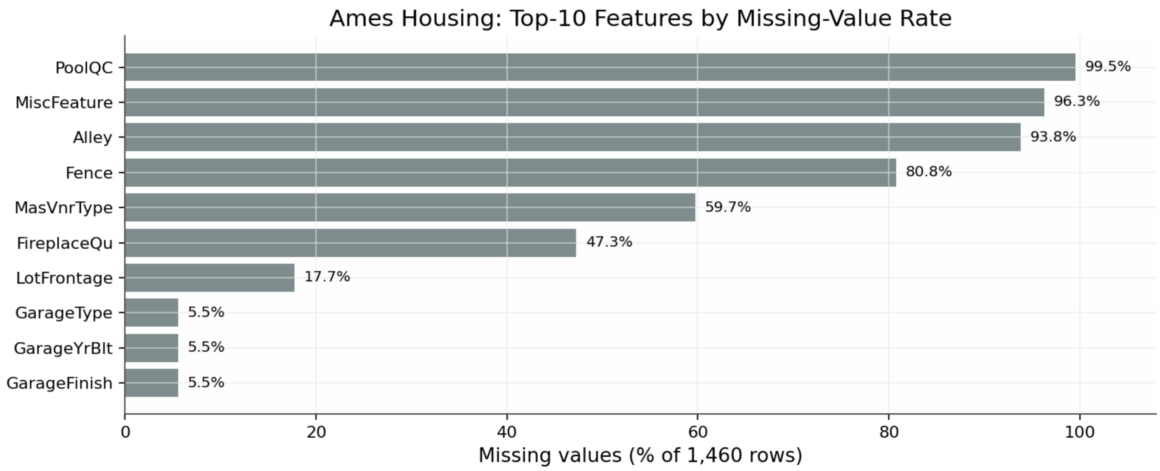

Figure 2. Ames Housing: top-10 features by missing-value rate. High rates reflect structural absence, not data corruption.

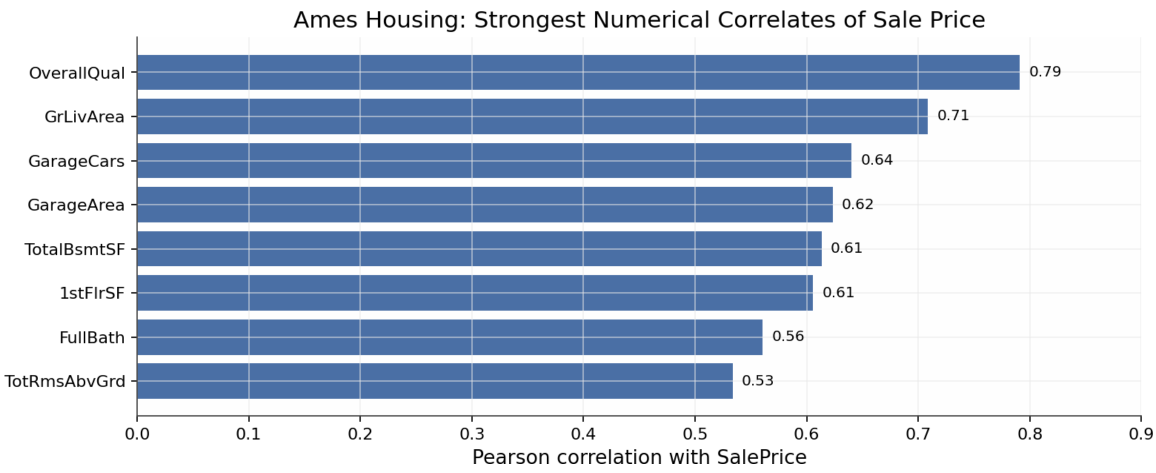

Figure 3. Ames Housing: strongest numerical correlates of SalePrice (Pearson r, full dataset).

5.2 California Housing

- **Right-censored target.** Figure 4 shows a sharp spike at $500,000: 992 of 20,640 block groups (4.8%) sit exactly at the cap, so the target is a *censored median*. Errors at the top of the market are therefore understated, and no model can recover the true value of a capped district. That is a ceiling on accuracy that comes from the data, not from the model.
- **Income saturation.** Figure 5 (left) shows district value climbing steeply with median income at the low end and then flattening out, with spread widening along the way. A straight line through this cloud has to compromise somewhere, which is the direct explanation for the linear family's RZ ceiling near 0.58 in Section 7.
- **Strong spatial structure.** Figure 5 (right) plots value by latitude and longitude. High values hug the coast — the Bay Area, Los Angeles/Orange County, San Diego — and drop off inland. Neither coordinate means much on its own; it is the interaction that matters, and that is the kind of pattern tree ensembles pick up and additive linear models cannot.

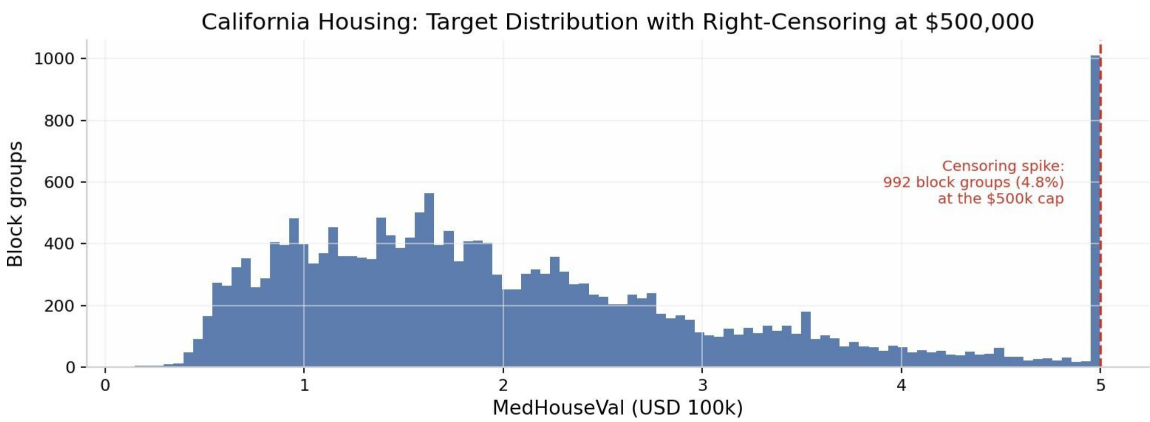

Figure 4. California Housing: target distribution. The spike at the right edge is right-censoring at the $500k cap (992 block groups, 4.8%).

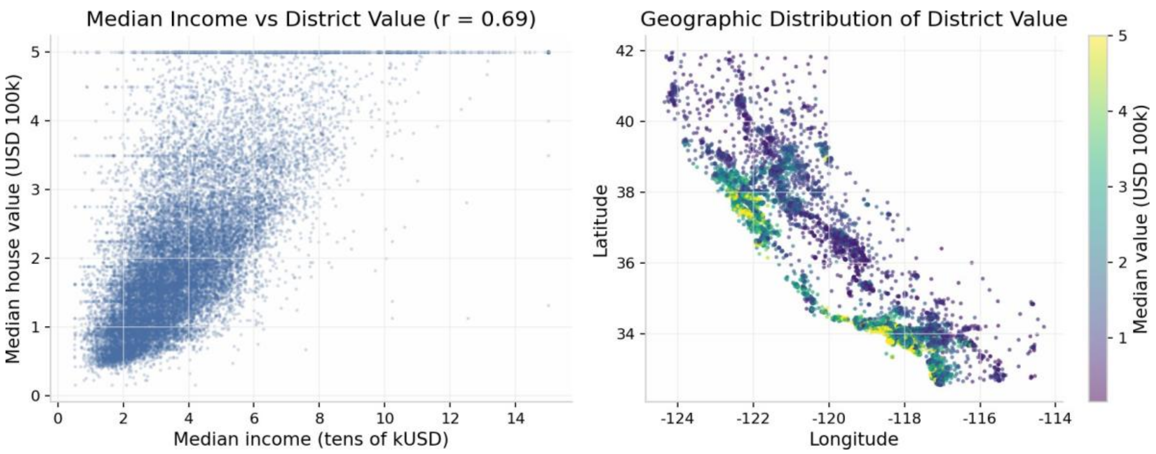

Figure 5. California Housing: median income vs district value showing saturation (left); geographic value surface showing coastal concentration (right).

5.3 UCI Real Estate Valuation

- **Log-linear transit premium.** Figure 6 (left) plots unit price against MRT-station distance on a log scale. The relationship is close to linear in log-distance (r = −0.67) and is the single strongest in the dataset. A linear model on raw distance cannot follow this curvature; logging the distance is an obvious feature-engineering fix.
- **Amenity and location effects.** Convenience-store count tracks price positively (r = 0.57): mean unit price climbs from about 23 to 43 (10k NTD/ping) as store count increases (Figure 6, right). Latitude and longitude add further spatial signal (r ≈ 0.55 and 0.52).
- **Mild skew and clean records.** The target is only mildly right-skewed (skewness 0.60; median 38.5; maximum 117.5), with no missing values and no duplicates. Given how clean this dataset is, it is not surprising that the four linear-family variants land almost on top of each other in Section 7.

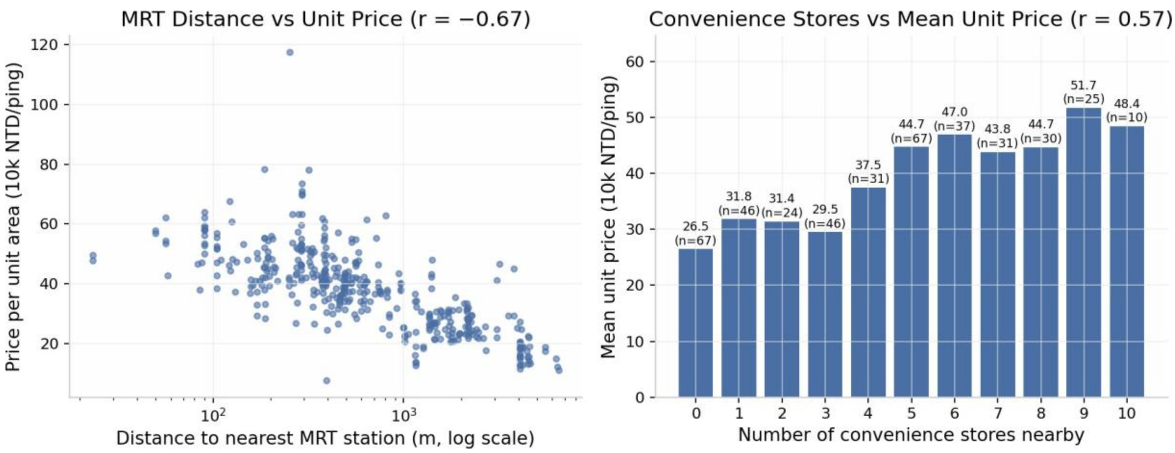

Figure 6. UCI Real Estate: unit price vs MRT-station distance on a log-distance scale (left); mean unit price by nearby convenience-store count (right).

6. Model Development

The experiments follow the lab's design matrix: a dummy mean baseline (E0); simple Linear Regression on one justified feature (E1 — GrLivArea for Ames, MedInc for California, MRT distance for UCI); multiple Linear Regression through the full pipeline (E2); the regularized variants (E3); a depth-limited Decision Tree and a 300-tree Random Forest (E4); Gradient Boosting (E5); the same battery rerun on all three datasets (E6); and a grid search over Ridge α as the controlled tuning experiment (E7).

The main settings were: Decision Tree with `max_depth=6` ; Random Forest with `n_estimators=300` and `min_samples_leaf=2` ; Gradient Boosting left at scikit-learn defaults; Lasso and Elastic Net at α = 0.001 with `max_iter=20000` . Since every model saw the same pipeline and the same split, the differences in the tables below come from the estimators themselves, not from how the data was handled.

7. Evaluation Results

7.1 Ames Housing (target: SalePrice, USD)

Gradient Boosting posted the lowest cross-validation RMSE of the seven candidates ($27,709) and held that lead on the test set (Table 2, Figure 7). The linear family bunches tightly at RZ ≈ 0.88–0.90, while the lone Decision Tree trails everyone, linear models included.

Table 2. Ames Housing — test-set comparison (292 held-out sales) and 5-fold CV on training data

Model	Test MAE ($)	Test RMSE ($)	Test R^2	CV RMSE ($)	CV R^2
Gradient Boosting	**16,555.67**	**26,503.77**	**0.9084**	**27,708.84**	**0.8668**
Lasso	18,010.80	28,337.93	0.8953	36,042.14	0.7747
Random Forest	17,473.41	28,747.70	0.8923	30,637.61	0.8400
Linear Regression	18,287.70	29,473.87	0.8867	36,396.35	0.7700
Elastic Net	18,809.63	29,709.27	0.8849	33,588.56	0.8057
Ridge	19,004.41	29,841.82	0.8839	32,817.04	0.8147
Decision Tree	26,729.14	39,928.28	0.7922	44,010.85	0.6716

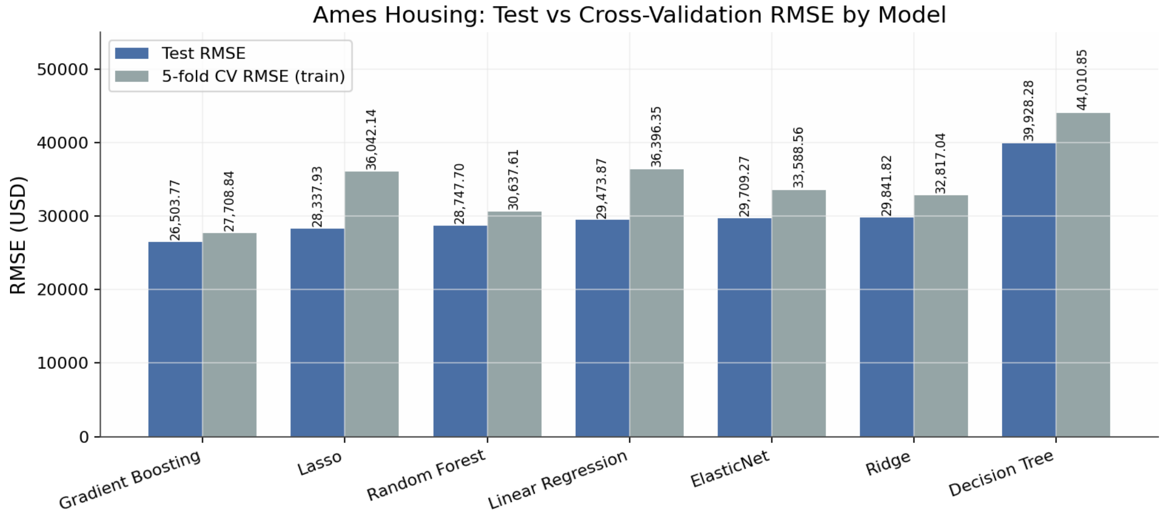

Figure 7. Ames Housing: test vs 5-fold CV RMSE. Gradient Boosting leads on both; the CV–test gap is small, indicating stable selection.

7.2 California Housing (target: median block-group value, USD 100k)

Random Forest won the cross-validation stage and confirmed it on test (Table 3, Figure 8). What stands out here is the split between model families. Linear Regression and Ridge stall at RZ ≈ 0.576, Lasso at α = 0.001 shrinks all the way back to the mean predictor (RZ ≈ 0.000), and the two ensembles clear RZ 0.80. A gap that size can only come from structure the linear models cannot see — the nonlinear income and geography effects documented in Section 5.2.

Table 3. California Housing — test-set comparison (4,128 held-out block groups) and 5-fold CV (units: USD 100,000)

Model	Test MAE	Test RMSE	Test R^2	CV RMSE	CV R^2
Random Forest	**0.3274**	**0.5051**	**0.8053**	**0.5109**	**0.8047**
Gradient Boosting	0.3717	0.5422	0.7756	0.5321	0.7881
Decision Tree	0.4539	0.7028	0.6230	0.7333	0.5973
Ridge	0.5332	0.7456	0.5758	0.7205	0.6115
Linear Regression	0.5332	0.7456	0.5758	0.7205	0.6115
Elastic Net	0.8060	1.0219	0.2031	1.0288	0.2080
Lasso	0.9061	1.1449	−0.0002	1.1562	−0.0002

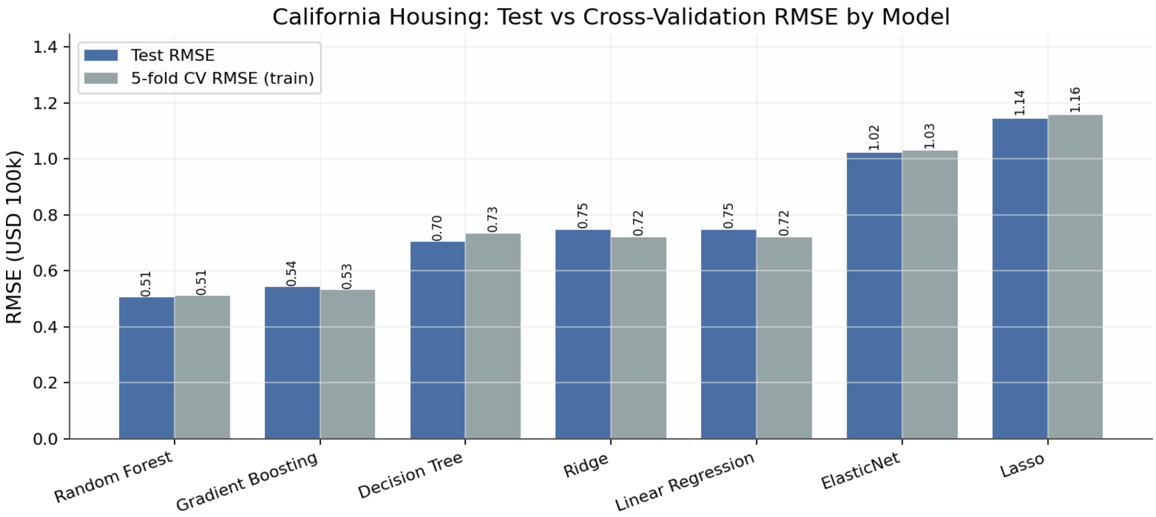

Figure 8. California Housing: test vs CV RMSE. Ensembles dominate; Lasso/Elastic Net collapse to the mean baseline at $\alpha = 0.001$.

7.3 UCI Real Estate Valuation (target: price per unit area, 10k NTD/ping)

Random Forest was again the cross-validation winner (Table 4, Figure 9). Gradient Boosting did edge it on test MAE (3.904 against 3.913), but it lost on test RMSE and was clearly worse in cross-validation. The selection rule was fixed in advance — choose by cross-validated training performance, then look at the test set once — so Random Forest is the model that gets selected, and the small test-MAE difference does not change that.

Table 4. UCI Real Estate — test-set comparison (83 held-out transactions) and 5-fold CV (units: 10,000 NTD per ping)

Model	Test MAE	Test RMSE	Test R^2	CV RMSE	CV R^2
Random Forest	**3.9131**	**5.6689**	**0.8084**	**7.8302**	**0.6643**
Gradient Boosting	3.9036	5.8443	0.7964	8.3502	0.6195
Ridge	5.3022	7.3108	0.6814	9.1083	0.5481
Elastic Net	5.3048	7.3139	0.6811	9.1105	0.5478
Lasso	5.3052	7.3144	0.6811	9.1108	0.5478
Linear Regression	5.3054	7.3148	0.6811	9.1109	0.5478
Decision Tree	5.9313	8.1530	0.6038	10.7388	0.3561

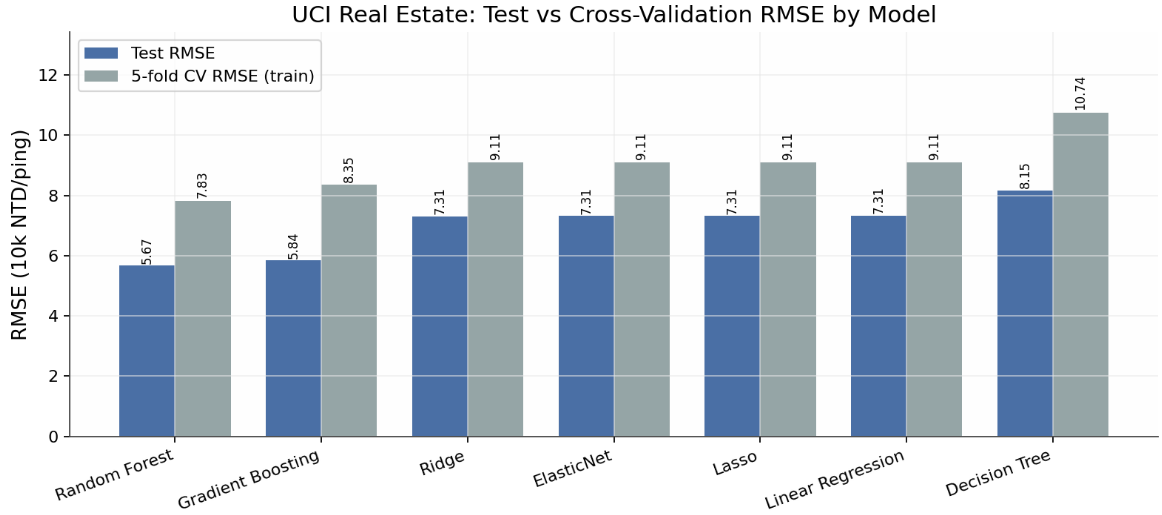

Figure 9. UCI Real Estate: test vs CV RMSE. The linear family is nearly identical across all four variants on this small dataset.

7.4 Cross-dataset synthesis

Figure 10 puts the three selected models side by side, and a few patterns repeat. The ensemble gain over the linear baseline scales with how nonlinear the problem is: +0.23 RZ on California, +0.13 on UCI, and only +0.02 on Ames. The CV–test gap stays small on the two larger datasets, which says the selection was stable. UCI is the odd one out: test RZ (0.808) comes in well above CV RZ (0.664). With 414 rows this is best read as small-sample luck on the test split rather than leakage, and the CV number is the safer estimate of what the model would do on new data.

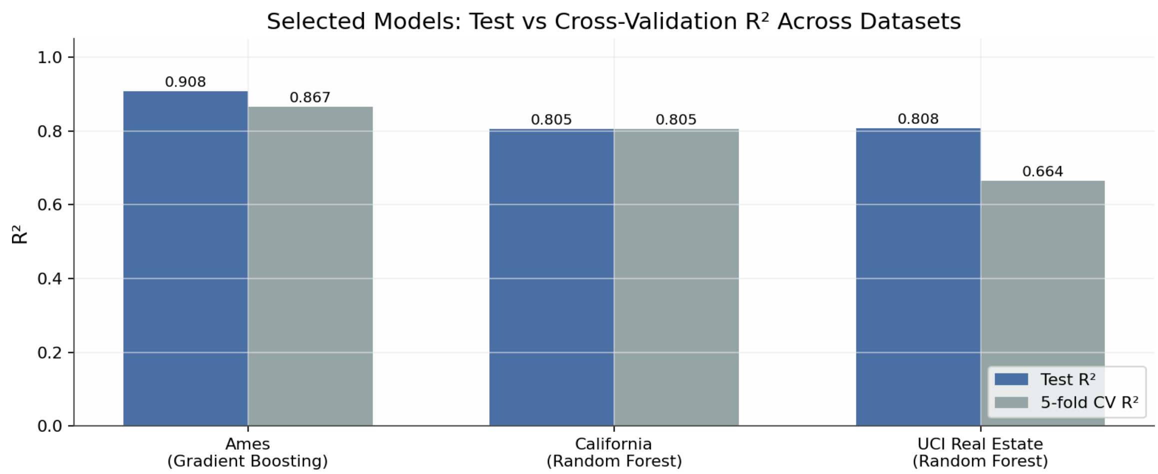

Figure 10. Selected models: test vs cross-validation RZ across the three datasets.

> **Adjusted R^2 caveat (Ames).** The adjusted RZ values for Ames are large negative numbers, which looks alarming but is an artifact, not a model failure. After one-hot encoding the design matrix has more features (p) than test observations (n = 292), so the penalty term (n−1)/(n−p−1) goes negative and drags the statistic with it. Adjusted RZ only means something when n ≫ p; in this regime cross-validation is the right way to compare models.

8. Interpretation

8.1 What drives price

Everything in this section is a conditional association, not a causal claim.

- **Ames.** Above-ground living area (GrLivArea), overall material and finish quality (OverallQual), neighborhood, and garage capacity top both the linear coefficient rankings and the tree-based importances. That the linear baseline already reaches test RZ 0.887 says most of the variance is additive. The roughly two extra RZ points from Gradient Boosting come from what the additive model cannot express — interactions such as quality × area, and diminishing returns at the expensive end.
- **California.** Median income dominates, with location (latitude/longitude, effectively coastal proximity), average rooms, and house age behind it. The 23-point RZ gap between the linear and ensemble models is what strong nonlinearity looks like in a metric.
- **UCI.** Distance to the nearest MRT station, convenience-store count, the two coordinates, and house age drive unit prices. Houses near rapid transit carry a premium, which matches standard urban economics.

8.2 The linear baseline's role

Linear Regression wins nowhere, but it is not wasted effort. On Ames it sits within about $3,000 RMSE of the best model, and every coefficient in it can be read and defended. On UCI it lands almost exactly on the regularized variants, which tells us multicollinearity is not a problem in that feature space. On California it fails — and the failure is informative, because it measures how much structure is nonlinear. That double duty, baseline and diagnostic, is the role it was kept in the study for.

8.3 Regularization behavior

Ridge never fell meaningfully below plain OLS, and on Ames it improved cross-validation RMSE by about $3,600 — a modest variance reduction in a design matrix where one-hot encoding creates many correlated columns. Lasso and Elastic Net behaved sensibly on Ames and UCI, then collapsed to the mean predictor on California at $\alpha = 0.001$. The lesson is that a useful α on one dataset can be a destructive one on another; the range has to be found by cross-validation each time, not carried over.

8.4 Generalization behavior

The two selected models generalize cleanly: test metrics match or slightly beat their cross-validation counterparts (Ames: test RMSE $26,504 against CV $27,709; California: 0.505 against 0.511), so there is no sign of overfitting at the selection stage. UCI runs in the opposite direction, with test above CV. That direction is the harmless one — it reflects fold-to-fold variance over 331 training rows, not leakage — but it is also a reminder that on small datasets the CV estimate comes with real uncertainty and should be reported as such.

9. Limitations and Risks

- **Historical drift.** All three models are fitted on past markets (Ames, 2006–2010; California, 1990; Taiwan, 2012–2013). Inflation, interest-rate cycles, and regime changes will erode absolute price predictions over time; the models capture relative structure more reliably than current price levels.
- **Target semantics.** California predicts a *median block-group* value censored at $500,001, not an individual home price, and UCI predicts price *per unit area*, not total price. Reading either output as an individual-property valuation would be a category error.
- **Random-split realism.** A random holdout assumes deployment on similar properties from the same market and period. A production setting would call for temporal or spatially grouped splits to get a genuinely forward-looking estimate.
- **Small-sample uncertainty (UCI).** With 83 test observations, a handful of transactions moves RZ by several points. The cross-validation estimate (0.664) is the figure to anchor expectations to.
- **Fairness and governance.** Neighborhood and coordinate features can encode socioeconomic and historical inequalities. Use in credit, taxation, or access decisions requires governance review, segment-level error analysis, and human escalation rules for atypical properties.
- **Unmeasured drivers.** None of the datasets records financing conditions, renovation history beyond year-remodeled, school quality, or micro-location attributes, and that caps the accuracy any model can reach here.
- **Limited tuning budget.** Only Ridge received a full grid search; the ensembles ran with fixed, defensible settings. Tuning learning rate, depth, and subsampling would likely close some of the remaining gap between the leaders.

10. Future Improvements

- **Log-transformed target.** The Ames skew (1.88) makes a `TransformedTargetRegressor` with log1p worth trying, since the high-end tail currently drives much of the error. Errors would still be reported after inverse transformation, in dollars.
- **Broader hyperparameter search.** Randomized or Bayesian search over the Gradient Boosting and Random Forest settings, with the compute budget stated up front.
- **Spatial and temporal validation.** Neighborhood-grouped or year-based splits would better approximate deployment, and engineered spatial features (e.g., distance to urban centers) could help on California.
- **Uncertainty quantification.** Quantile regression or conformal intervals would attach an error range to every estimate, which is what professional valuation practice expects.
- **Segment-level error analysis.** Scoring errors by price band and neighborhood would show where the models struggle — very expensive or very old properties are the likely candidates.
- **Deployment packaging.** The artifacts are already serialized with `joblib` ; what remains is input-schema validation, drift monitoring, and scheduled retraining.

11. Conclusion

The study met its success criteria on all three datasets. Each final model cuts RMSE by 56–70% against the mean baseline, the CV–test gaps on the two larger datasets are small, and the whole pipeline is reproducible from a fixed seed with recorded versions and serialized artifacts. As for the question the lab poses — how accurately, reliably, and transparently can property value be predicted — the answer depends on the data. On Ames, where the signal is largely additive, Gradient Boosting reaches RZ = 0.908 while the linear baseline stays within striking distance and remains fully interpretable. On California and UCI the structure is genuinely nonlinear, and the tree ensembles are the only models that capture it. Linear Regression earns its place either way: as a competitive, transparent baseline where additivity holds, and as a diagnostic that quantifies nonlinearity where it does not. Within the markets and periods they were trained on, the selected models are fit for decision support. Outside that coverage — today's markets, individual California homes, or high-stakes credit decisions — they need revalidation, uncertainty estimates, and human oversight first.

Appendix A. Environment, Artifacts and Reproducibility

Table 5. Execution environment (from run_metadata.json; executed 2026-07-19)

Component	Version / setting	Component	Version / setting
Python	3.11.0	scikit-learn	1.8.0
numpy	1.26.4	matplotlib	3.10.8
pandas	2.3.3	seaborn	0.13.2
joblib	1.5.3	Random seed	42 (all splits, CV, stochastic models)

Table 6. Submitted artifacts per dataset

Dataset	Selected model	Model artifact (joblib)	Result CSVs
Ames Housing	Gradient Boosting	`ames_house_price_pipeline.joblib`	`ames_model_comparison.csv`, `ames_cross_validation.csv`
California Housing	Random Forest	`California_house_price_pipeline.joblib`	`California_model_comparison.csv`, `California_cross_validation_results.csv`
UCI Real Estate	Random Forest	`uci_real_estate_model.joblib`	`uci_model_comparison.csv`, `uci_cross_validation.csv`

Supporting materials include the three analysis notebooks (Ames, California, UCI), each self-contained; `23MID0043_Lab01_README.md` with the execution order and environment setup; `execute_notebooks.py` , a small harness that reruns the notebooks headlessly; and `run_metadata.json` , which logs the seed, split sizes, selected models, and headline metrics. Each notebook runs top to bottom from a fresh kernel, and the raw data files are never modified.

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